

# Strongly Coupled Partitioned FSI Solver in MFEM with Scalable Interface Transfer

L. Molinari<sup>1,2</sup>, D. Serino<sup>1</sup>, J. Andrej<sup>4</sup>, Q. Tang<sup>3</sup>

<sup>1</sup>Los Alamos National Laboratory, Los Alamos, NM, USA <sup>2</sup>Department of Mathematics, Emory University, Atlanta, GA <sup>3</sup>School of Computational Science and Engineering, Georgia Institute of Technology, Atlanta, GA, USA <sup>4</sup>Lawrence Livermore National Laboratory, Livermore, CA, USA

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## Introduction & Motivation

Fluid-Structure Interaction (FSI) is crucial for modeling complex coupled phenomena across engineering and biomedical fields. While versatile, partitioned FSI schemes often struggle with stability and computational cost, especially due to the “added mass” effect.

### ■ Goal:

- Develop robust framework for partitioned FSI problems (incompressible Navier-Stokes (ALE), linear elasticity).
- Scalable and robust `gslib`-based interface transfer methods.
- Classical strong coupling methods and investigation of stable matching condition for subiteration-free partitioned FSI scheme.

## Mathematical Model

### Incompressible Navier-Stokes Equations (INS) [1]

- Formulated in **Arbitrary Lagrangian-Eulerian (ALE)** framework.
- **BDF** time discretization, **implicit-explicit time-integration (IMEX)** for viscous/convective terms
- Pressure-Projection method for decoupling INS solution:

$$\begin{aligned} \Delta p &= \rho \nabla \cdot (L_{\times}^*(\mathbf{u}) + F^*(\mathbf{u})) && \text{in } \Omega_{fluid} \\ \nabla p \cdot \mathbf{n} &= -\frac{b_0}{\Delta t} \mathbf{u}_D \cdot \mathbf{n} + (L_{\times}^*(\mathbf{u}) + F^*(\mathbf{u})) \cdot \mathbf{n} && \text{on } \Gamma_D \\ \rho \beta p + \nabla p \cdot \mathbf{n} &= \rho [(L_{\times}(\mathbf{u}) + F^*(\mathbf{u})) \cdot \mathbf{n} + \beta(n \cdot \boldsymbol{\tau}) \cdot \mathbf{n} - \frac{b_0}{\Delta t} \mathbf{u}_s \cdot \mathbf{n} - \beta(\boldsymbol{\sigma}_s \cdot \mathbf{n}) \cdot \mathbf{n}] && \text{on } \Gamma_R \\ \frac{b_0}{\Delta t} \mathbf{u} - L(\mathbf{u}) &= F^*(\mathbf{u}) - \frac{1}{\rho} \nabla p && \text{in } \Omega_{fluid} \\ \mathbf{u} &= \mathbf{u}_D && \text{on } \Gamma_D \\ \alpha \mathbf{u} + \boldsymbol{\sigma}_f \cdot \mathbf{n} &= \alpha \mathbf{u}_s + \boldsymbol{\sigma}_s \cdot \mathbf{n} && \text{on } \Gamma_R \end{aligned} \quad (1)$$
$$L(\mathbf{u}) = \nabla \cdot [\boldsymbol{\tau}], L_{\times}(\mathbf{u}) = -\nabla \times \nabla \times \mathbf{u}, \boldsymbol{\tau} = \nu(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \beta = \frac{1}{\alpha} \frac{b_0}{\Delta t}, F(\mathbf{u}) \text{ explicit terms.}$$

### Linear Elasticity

- Solved in the **reference configuration** using an **implicit HHT-alpha time integration** method.

$$\rho_s \frac{\partial^2 \mathbf{d}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma}_s + \mathbf{b} \quad \text{in } \Omega_{solid} \quad (2)$$

$$\boldsymbol{\sigma}_s = -\lambda(\nabla \cdot \mathbf{d})\mathbf{I} + \mu_s(\nabla \mathbf{d} + \nabla \mathbf{d}^T)$$

### Interface Coupling Conditions ( $\Gamma_{FSI}$ ):

- **Kinematic (Velocity Continuity):**  $\mathbf{u}_f = \mathbf{u}_s = \frac{\partial \mathbf{d}}{\partial t}$
- **Dynamic (Traction Continuity):**  $\boldsymbol{\sigma}_f \cdot \mathbf{n}_f + \boldsymbol{\sigma}_s \cdot \mathbf{n}_s = \mathbf{0}$

### Mesh Movement

- Implemented different mesh deformation algorithms [2]: **harmonic extension (HE)**, incremental version **IHE)**, **linear elasticity (LE)**, with prescribed solid displacement at the fluid-solid interface.

## FSI Partitioned Algorithm

- **Coupling Scheme:** Strongly coupled Dirichlet-Neumann (**DN**) and Robin-Neumann (**RN**) schemes (Fluid: D/R, Solid: N).
- **Dynamic Relaxation:** Aitken’s  $\Delta^2$  method

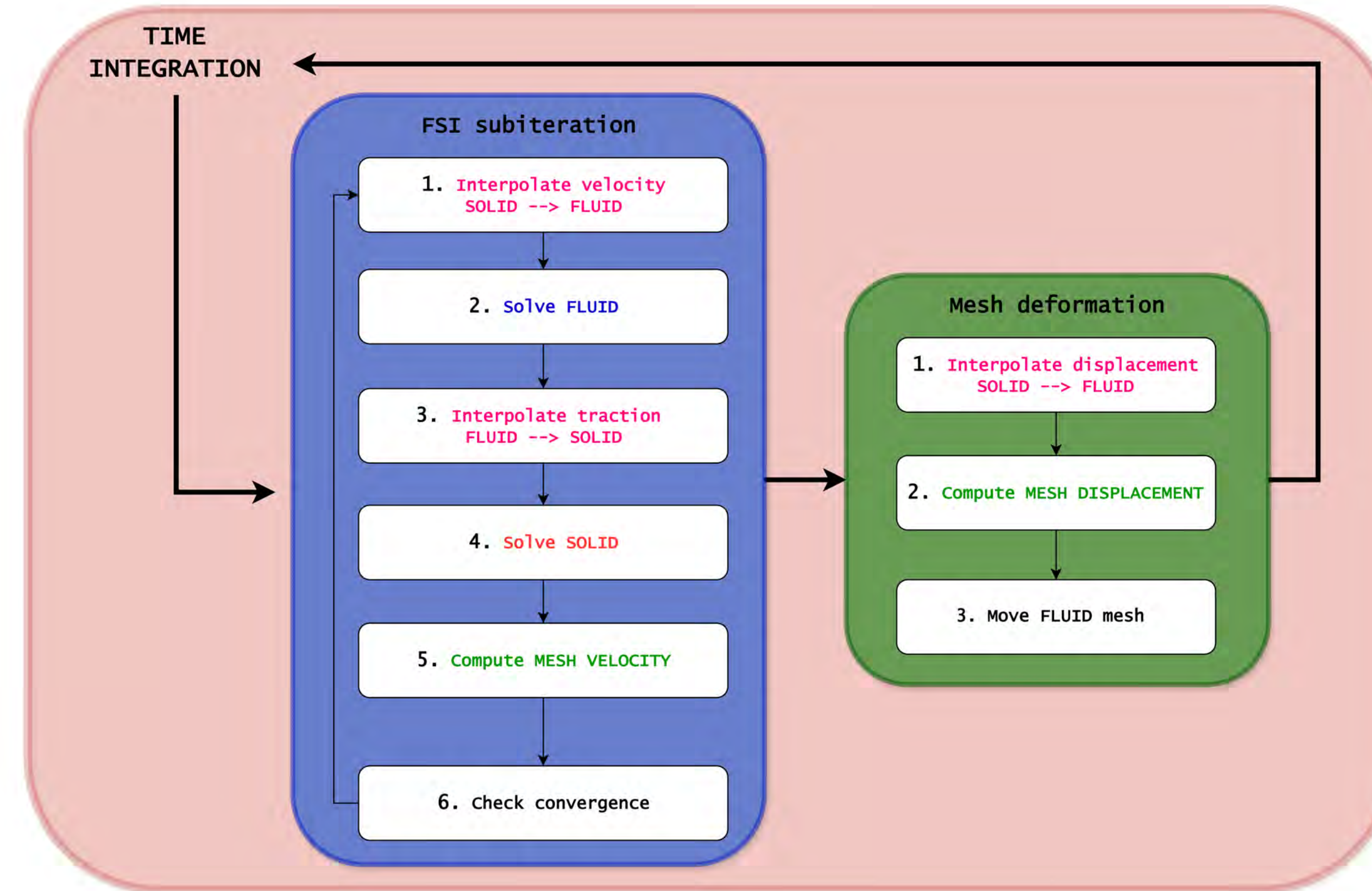


Figure: Workflow of the two-way FSI coupling and ALE mesh update process.

## Interface Transfer (`gslib`-based)

Flexible InterfaceTransfer class, built upon the `gslib` library. `gslib` is a library for Gather/Scatter-type nearest neighbor data exchanges for SEM, FEM, and FD [3].

- **Versatile Data Support:** Interpolates scalar/vector fields and custom quantities of interest (e.g., stress tensors).
- **Non-Conforming Interfaces:** non-matching interfaces and/or different FE spaces order.
- **Parallel Scalable implementation**
- **Moving Meshes:** Works with source/destination in different configurations.

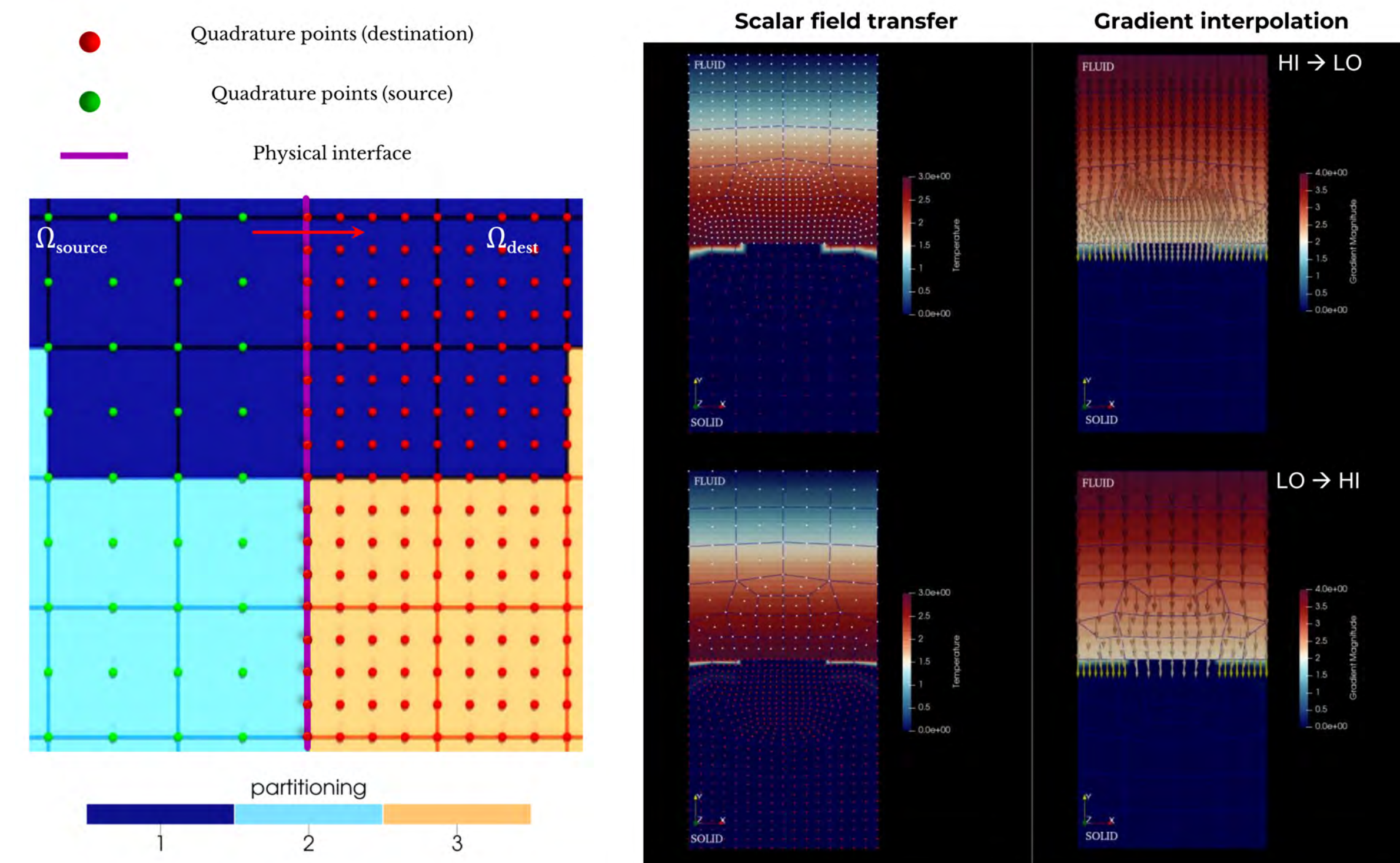


Figure: Scalar field and gradient transfer across non-matching interface.

## Results

### ■ Partitioned Solver Verification:

- **Fluid/Solid Solvers:** Analytic tests using Method of Manufactured Solutions (MMS), convergence studies, fluid/solid mechanics benchmarks.

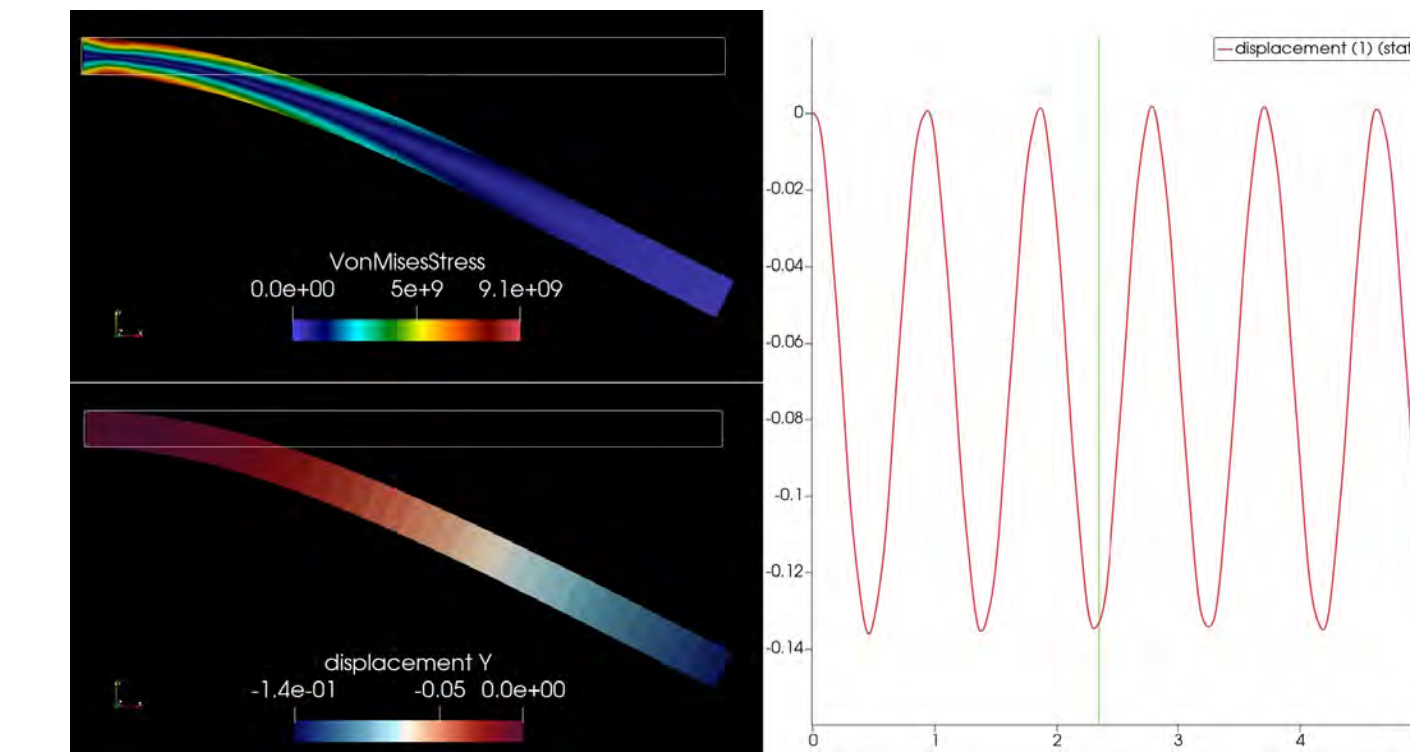


Figure: CSM benchmark [4].

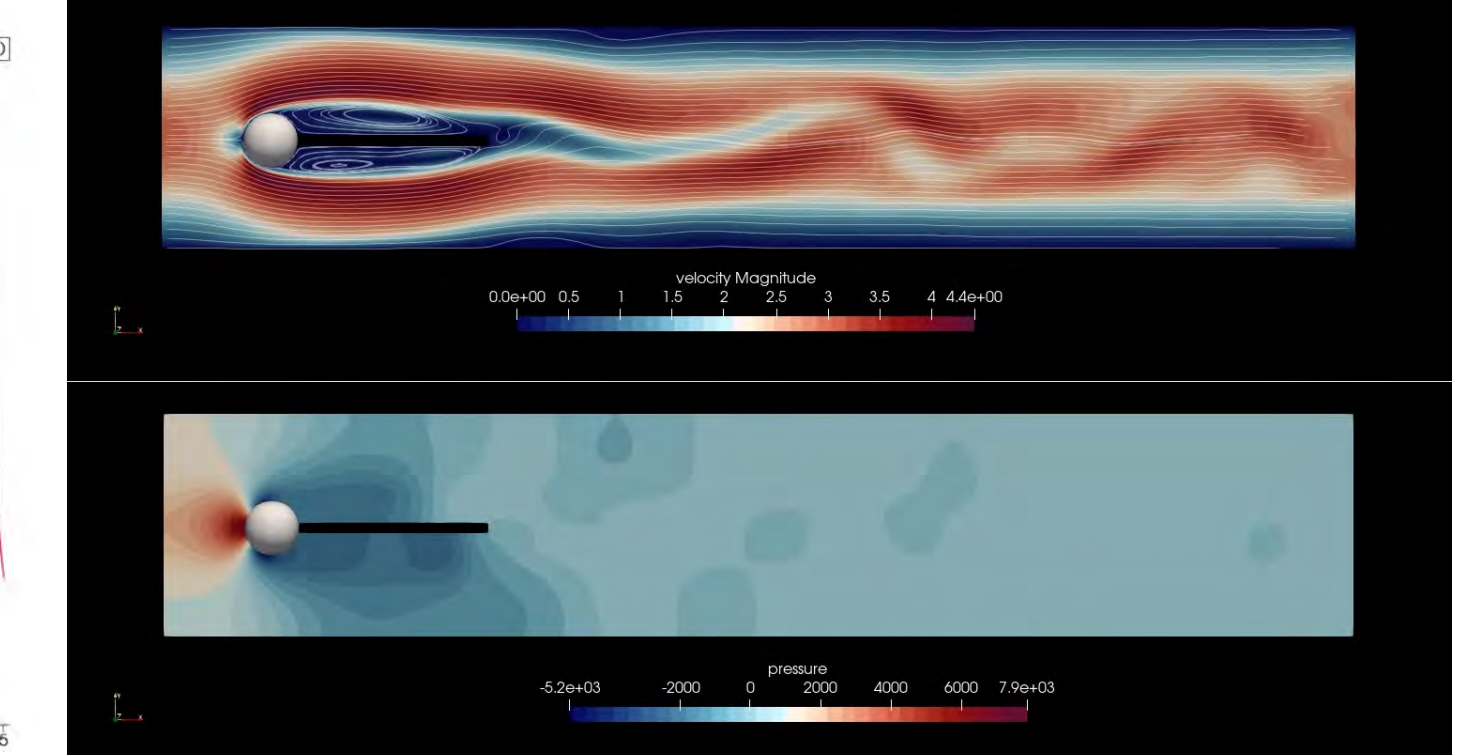


Figure: CFD3 benchmark [4],  $Re = 200$ .

### ■ FSI Test Cases: Solver capabilities demonstrated on different FSI benchmark problems:

- Turek Benchmark [4], Elastic Piston Test, Transverse Shear Test [5]

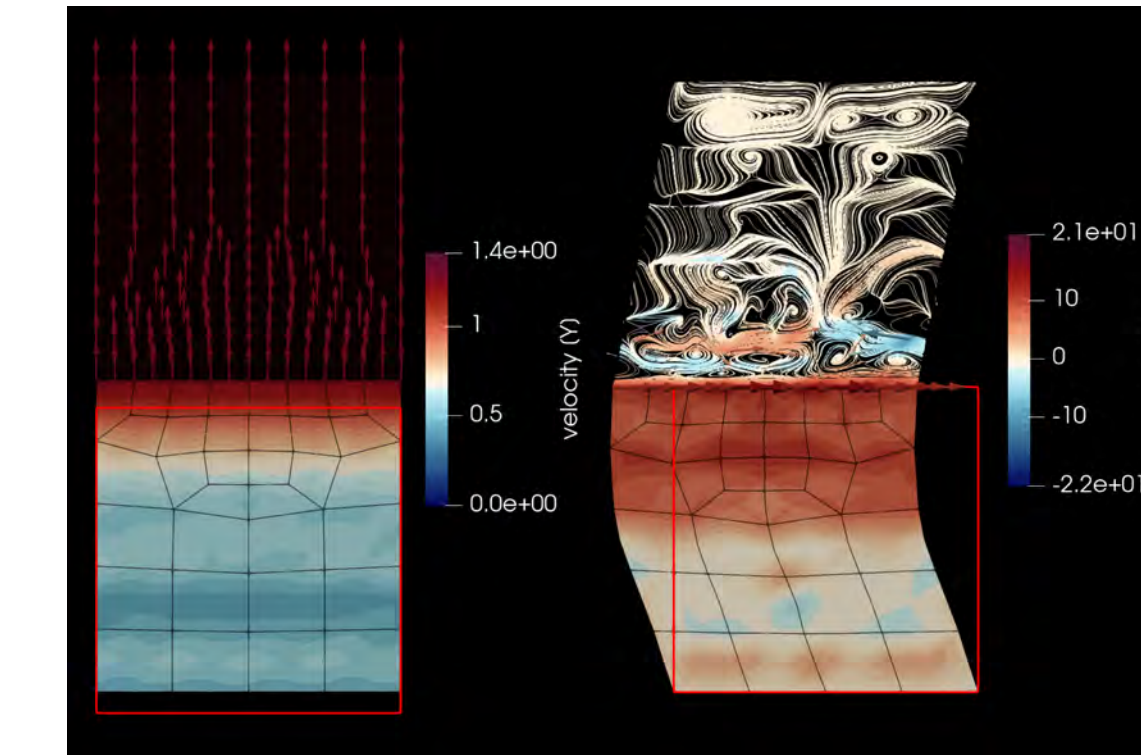


Figure: Piston and transverse test.

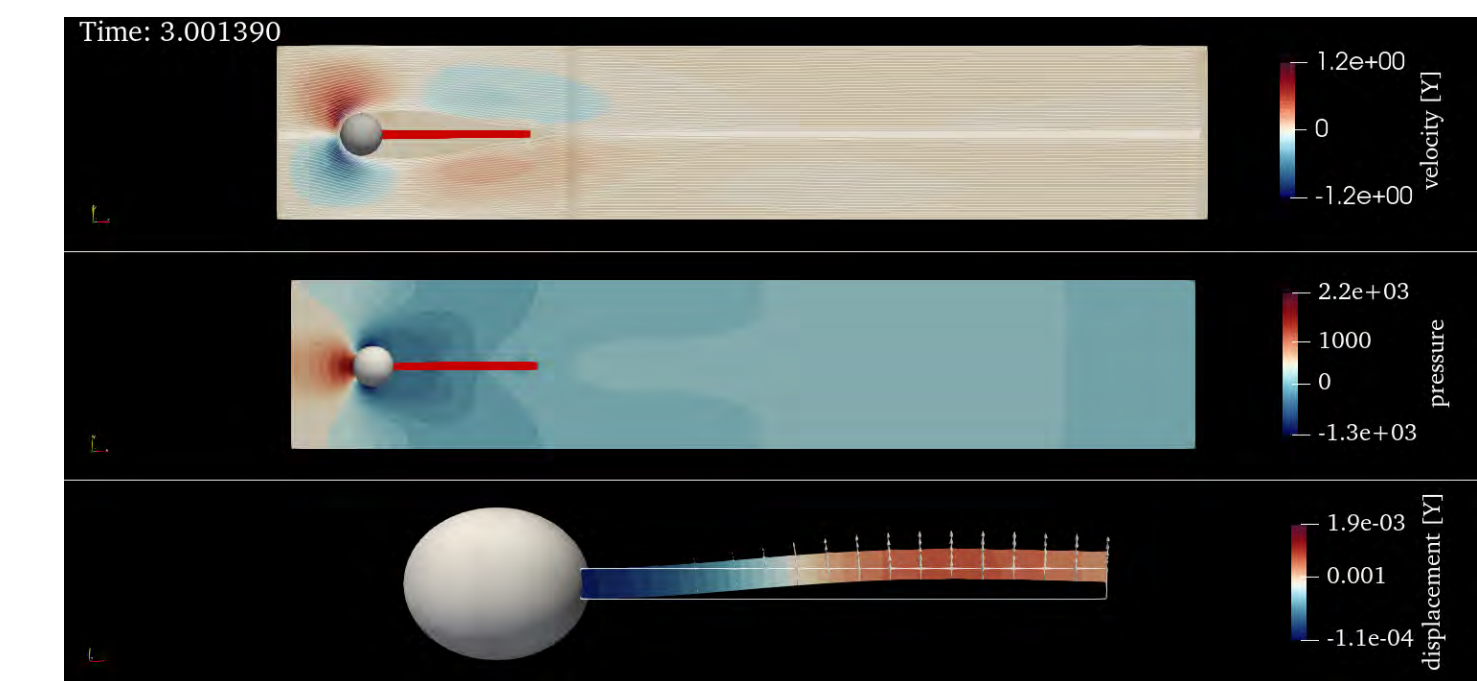


Figure: Turek (FSI2) benchmark [4].

### ■ Dynamic acceleration

- Improved convergence with dynamic relaxation for cases with significant added mass effect.

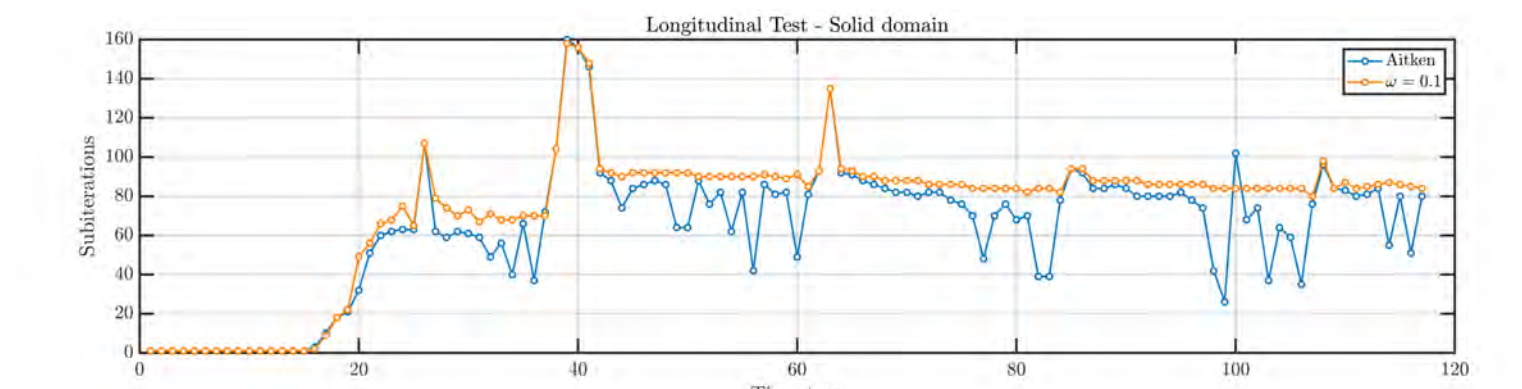


Figure: Dynamic vs static relaxation for longitudinal test.

## Conclusion

- Successfully developed a **strongly coupled partitioned FSI solver** in MFEM.
- Implemented a scalable, **`gslib`-based InterfaceTransfer class**, enabling efficient, high-order data exchange on complex and non-conforming interfaces.
- **Future Work:** We’re investigating specialized boundary conditions to mitigate the added mass effect, striving for a stable, subiteration-free FSI algorithm applicable to challenging problems like floating objects and free surface flows.

## References

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